

Anti-genotoxic effect of Ocimum sanctum L. extract against cyproterone acetate induced genotoxic damage in cultured mammalian cells.



The anti-genotoxic effect of *Ocimum sanctum* L. extract was studied against the genotoxic effect induced by a synthetic progestin cyproterone acetate, on human lymphocytes using chromosomal aberrations, mitotic index, sister chromatid exchanges and replication index as parameters. About 30 microM of cyproterone acetate was treated with *O. sanctum* L. infusion, at dosages of 1.075×10^{-4} , 2.125×10^{-4} and 3.15×10^{-4} g/ml of culture medium. A clear dose-dependent decrease in the genotoxic damage of cyproterone acetate was observed, suggesting a possible modulating role of the plant infusion. The results of the present study suggest that the plant infusion per se does not have genotoxic potential, but can modulate the genotoxicity of cyproterone acetate on human lymphocytes in vitro.